CNENGR 116
Fundamentals of Digital Electronics

Lecture 2 – Logic Gates & Boolean Algebra

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Overview

- Basic logic operations
- Basic logic gates
- Logic circuit diagrams
- Boolean Algebra (Laws and theorems)
- DeMorgan's Theorem
- Examples

Objectives

- Introduce basic concepts of logic operations
- Define NOT, AND, OR and XOR operations with associated truth tables and symbols
- Introduce Boolean algebra laws and theorems

Basic logic operations

- Logic in digital electronics deals with HIGH and LOW levels or 1 and 0 respectively.
- There are a few basic logic operations that form all other combinations: NOT, AND, OR and XOR*.
- The output of such logic operations is also a 1 or 0 state for a given input signals.
- This is shown in **truth tables** for each operation, which depict an **output** of the system for any combination of **inputs**

* Strictly speaking XOR can be represented as a combination of other 3

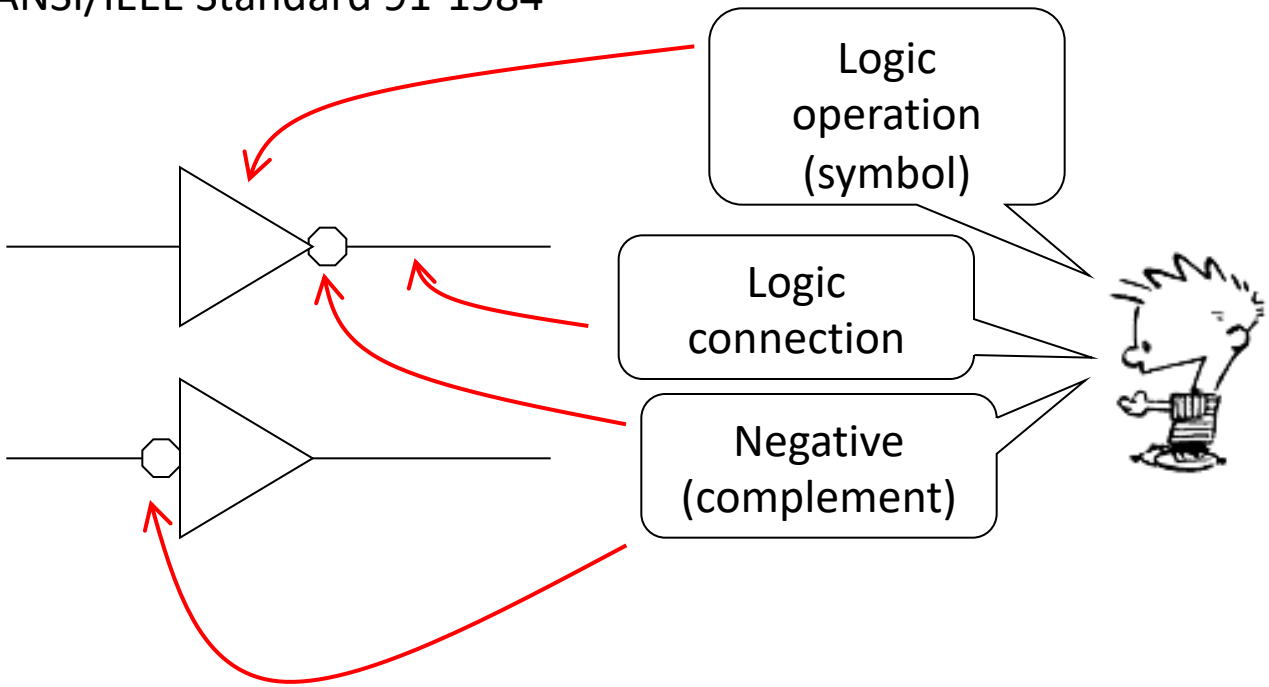
Definitions

- Variable in Boolean algebra is a symbol (usually an italic uppercase letter, eg:A) used to represent a logic quantity. Any single variable can have a 1 or 0 value.
- Complement is the inverse of a variable and is indicated by a bar over the variable ($\overline{A}$)
- Literal is a variable or a compliment of a variable

Introduction to logic gates

Logic symbols used to represent logic gates

in accordance with ANSI/IEEE Standard 91-1984



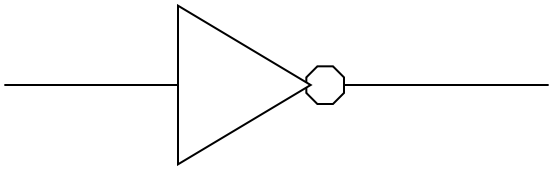
NOT

NOT operation performs an inversion operation. $X = \overline{A}$

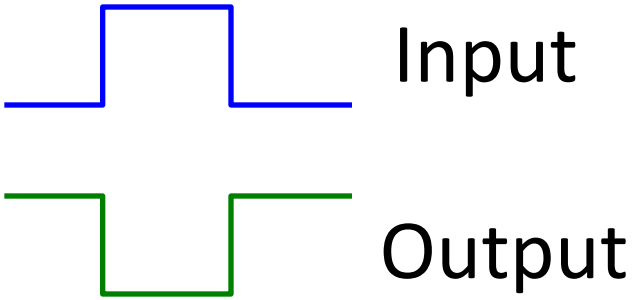
Truth Table

A	X
1	0
0	1

Symbol:



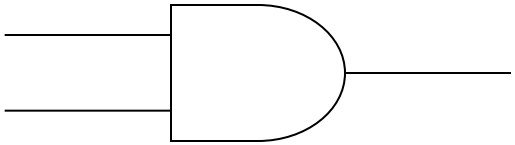
Timing Diagram:

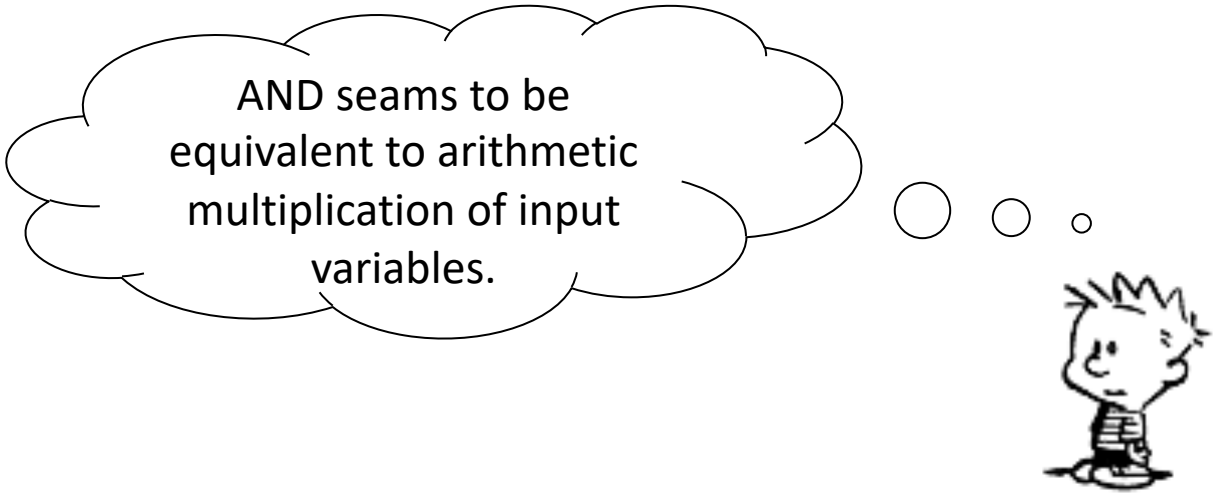


AND

AND operation requires all its inputs to be HIGH for the output to be HIGH.

Symbol:





Truth table for $X = A$ and B

A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

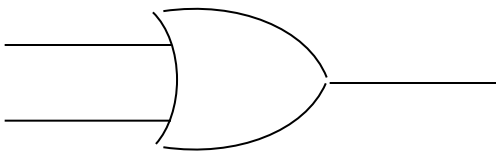
OR

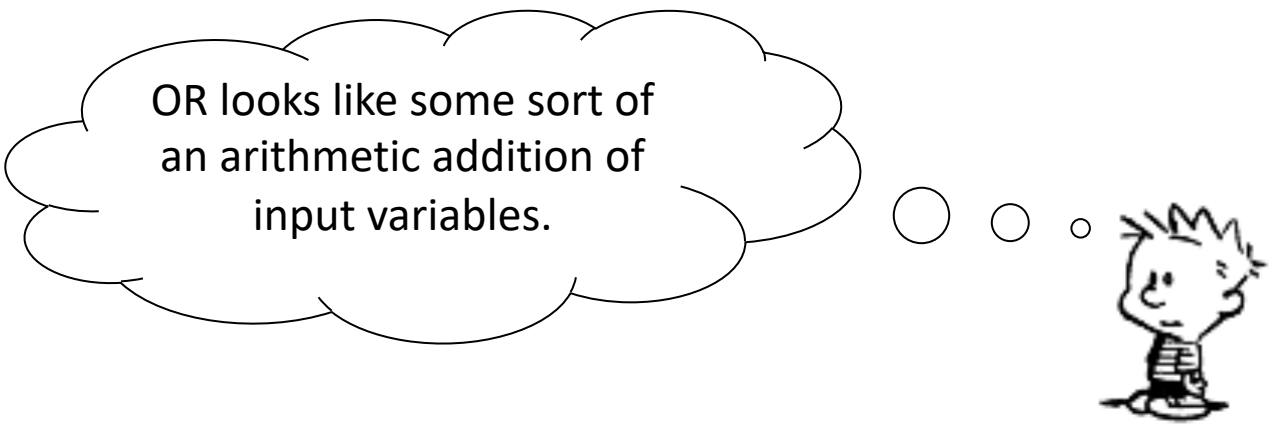
OR operation requires at least one of the inputs to be HIGH for the output to be HIGH.

Truth table for $X = A$ or B

A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

Symbol:

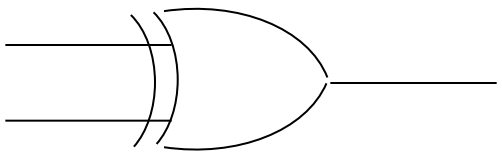




Exclusive-OR (XOR)

XOR operation requires only one of the inputs to be HIGH for the output to be HIGH.

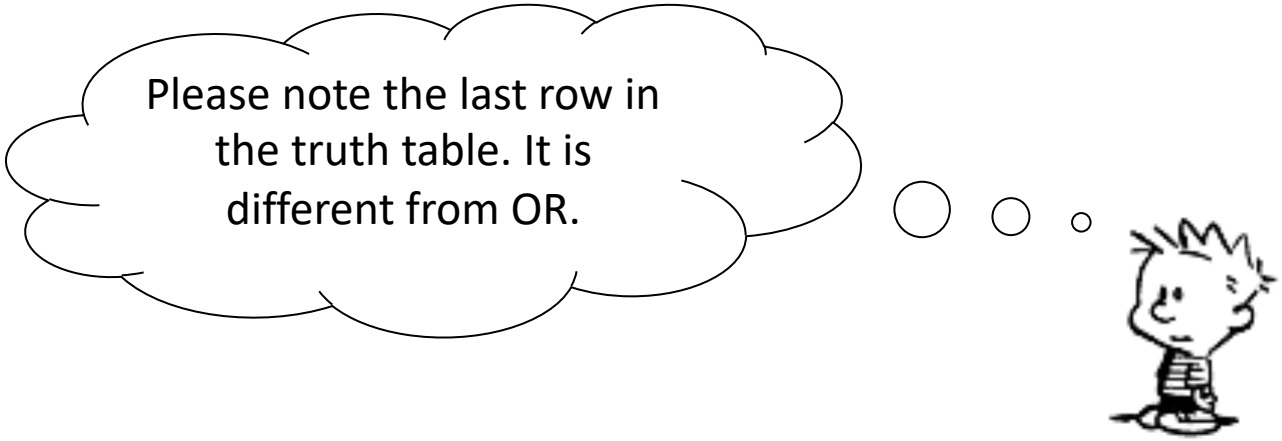
Symbol:



Truth table for

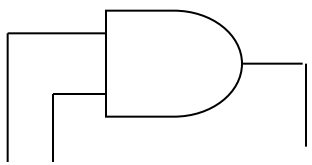
$X = A$ xor B

A	B	X
0	0	0
0	1	1
1	0	1
1	1	0



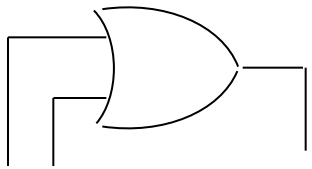
Graphical representation

AND



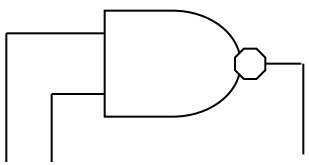
0 0	0
0 1	0
1 0	0
1 1	1

OR



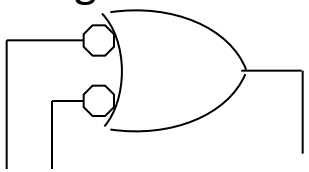
0 0	0
0 1	1
1 0	1
1 1	1

NAND



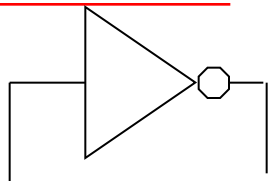
0 0	1
0 1	1
1 0	1
1 1	0

Negative-OR



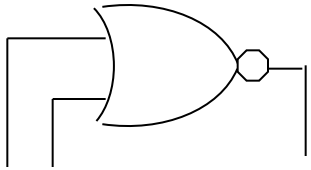
0 0	1
0 1	1
1 0	1
1 1	0

Inverter



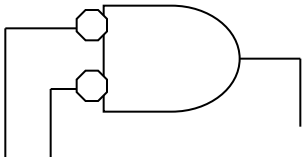
1	0
0	1

NOR



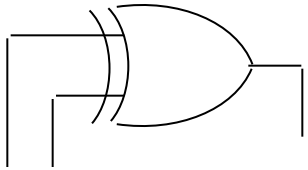
0 0	1
0 1	0
1 0	0
1 1	0

Negative-AND



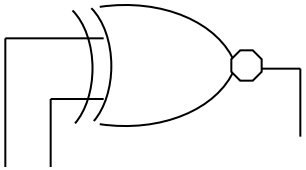
0 0	1
0 1	0
1 0	0
1 1	0

Exclusive-OR



0 0	0
0 1	1
1 0	1
1 1	0

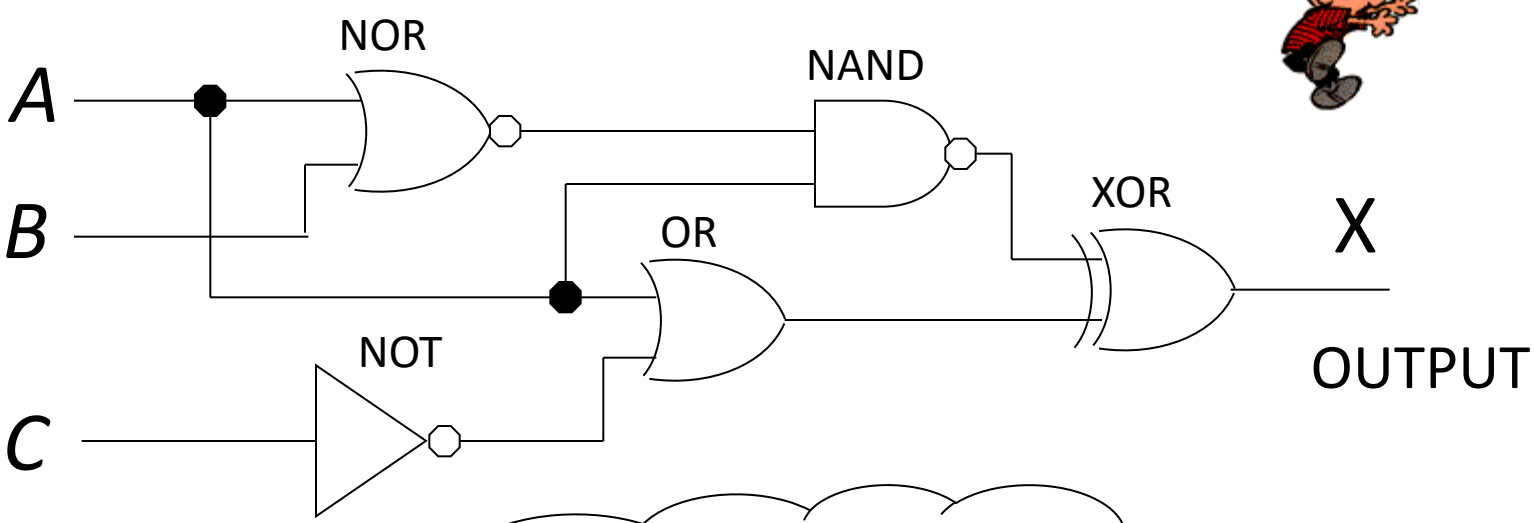
Exclusive-NOR

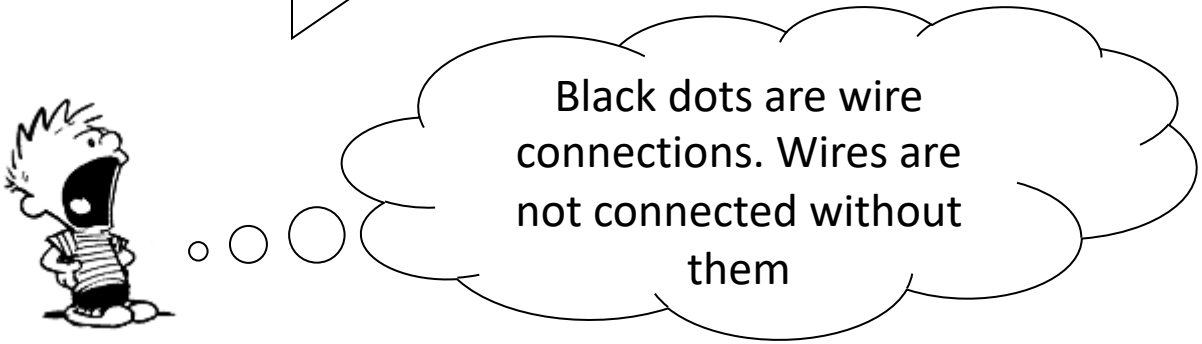


0 0	1
0 1	0
1 0	0
1 1	1

Logic circuit diagram

3 Variable input





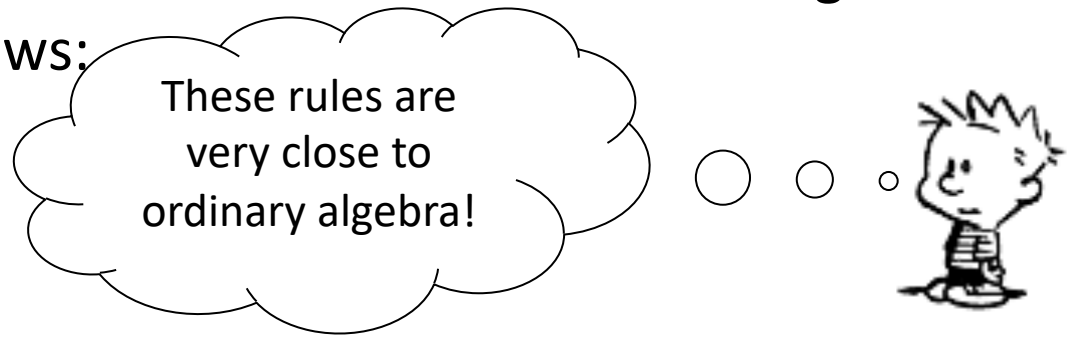
Boolean Algebra

Boolean Addition is equivalent to the OR operation and the basic rules are illustrated with relation to OR gate as follows:

$A + B = X$
$0 + 0 = 0$
$0 + 1 = 1$
$1 + 0 = 1$
$1 + 1 = 1$

Boolean Multiplication is equivalent to the AND operation and the basic rules are illustrated with relation to the AND gate as follows:

$A \bullet B = X$
$0 \bullet 0 = 0$
$0 \bullet 1 = 0$
$1 \bullet 0 = 0$
$1 \bullet 1 = 1$



Laws of Boolean Algebra

Commutative Laws

Addition: $A + B = B + A$

Multiplication: $AB = BA$

Associative Laws

Addition: $A+(B+C) = (A+B)+C$

Multiplication: $A(BC) = (AB)C$

Distributive Law $A(B+C) = AB + AC$

Rules of Boolean Algebra

Rules of Boolean algebra are used to manipulate and simplify Boolean expressions.

This has applications in the construction of simplified digital logic circuits and is thus important to us.

$1 \ldots A + 0 = A$

$2 \ldots A + 1 = 1$

$3 \ldots A \bullet 0 = 0$

$4 \ldots A \bullet 1 = A$

$5 \ldots A + A = A$

$6 \ldots A + \overline{A} = 1$

$7 \ldots A \bullet A = A$

$8 \ldots A \bullet \overline{A} = 0$

$9 \ldots \overline{\overline{A}} = A$

$10 \ldots A + AB = A$

$11 \ldots A + \overline{A}B = A + B$

$12 \ldots (A + B)(A + C) = A + BC$

Rule # 1 to 4

Rule #1: $A + 0 = A$

Rule #2: $A + 1 = 1$

Rule #3: $A \bullet 0 = 0$

Rule #4: $A \bullet 1 = A$

NB: These rules are similar to Ordinary arithmetic rules

A	**OR**	**0**	**=**	**A**
0	+	0	=	0
1	+	0	=	1

A	**OR**	**1**	**=**	**1**
0	+	1	=	1
1	+	1	=	1

A	**AND**	**0**	**=**	**0**
0	•	0	=	0
1	•	0	=	0

A	**AND**	**1**	**=**	**A**
0	•	1	=	0
1	•	1	=	1

Rule # 5 and 6

Rule #5: A + A = A

"Weather forecast - It is going to rain or it is going to rain"

A	OR	A	=	A
0	+	0	=	0
1	+	1	=	1

Rule #6: $A + \overline{A} = 1$

"Weather forecast - It is going to rain or it's not going to rain." Such a statement does not take any risks and is always true.

A	OR	$\overline{A}$	=	1
0	+	1	=	1
1	+	0	=	1

Rule # 7 and 8

Rule #7: $A \bullet A = A$

"I like going out AND I like going out" statement simply states that I like going out.

A	AND	A	=	A
0	•	0	=	0
1	•	1	=	1

Rule #8: $A \bullet \overline{A} = 0$

"It's raining and it's not raining". The statement will always be false.

A	AND	$\overline{A}$	=	0
0	•	1	=	0
1	•	0	=	0

Rule # 9

Rule #9: $\overline{\overline{A}} = A$

Double inversion action is the same as no action at all. Therefore an improper use of language such as “I have not done nothing” simply means that I HAVE done something rather than an even stronger negative statement.

A	NOT (A)	NOT (NOT (A)) = A
0	1	0
1	0	1

Rule # 10

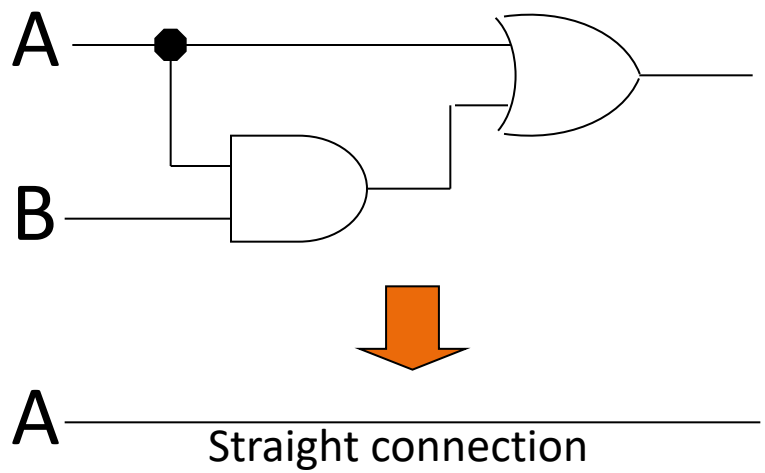
Rule #10: $A + AB = A$

$A + AB = A(1 + B)$ (Factorising)

$= A \bullet 1$ rule 2: $(1 + B) = 1$

$= A$ rule 4: $A \bullet 1 = A$

A	B	AB	A+AB
0	0	0	0
0	1	0	0
1	0	0	1
1	1	1	1



Rule # 11

$$A + \overline{A}B = A + B$$

Proof:

$$\begin{aligned} A + \overline{A}B &= (A + AB) + \overline{A}B \\ &= (AA + AB) + \overline{A}B \\ &= AA + AB + A\overline{A} + \overline{A}B \\ &= (A + \overline{A})(A + B) \\ &= 1 \bullet (A + B) \\ &= A + B \end{aligned}$$

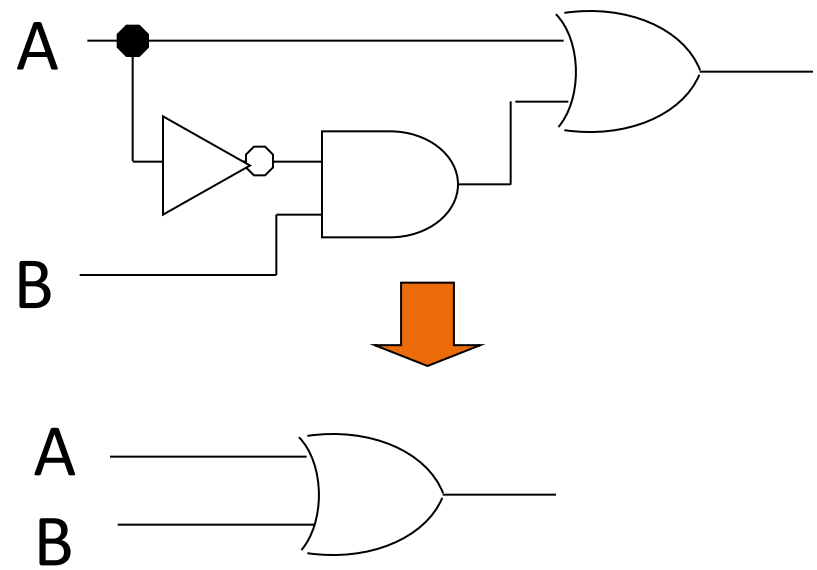
Rule #10: $A + AB = A$

Rule #7: $AA = A$

Rule #8: $A\bar{A} = 0$

Rule #6: $A + \bar{A} = 1$

A	B	$\overline{A}B$	$A + \overline{A}B$	$A+B$
0	0	0	0	0
0	1	1	1	1
1	0	0	1	1
1	1	0	1	1



Rule # 12 $(A + B)(A + C) = A + BC$

Proof: $(A + B)(A + C) = AA + AC + AB + BC$

$= A + AC + AB + BC$ *Rule #7: $AA = A$*

$= A(1 + C) + AB + BC$

$= A \bullet 1 + AB + BC$ *Rule #2: $A + 1 = 1$*

$= A(1 + B) + BC$ *Rule #4: $A \cdot 1 = A$*

$= A \bullet 1 + BC = A + BC$ *Rule #2: $A + 1 = 1$* *Rule #4: $A \cdot 1 = A$*

A	B	C	A+B	A+C	(A+B) (A+C)	BC	A+BC
0	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0
0	1	0	1	0	0	0	0
0	1	1	1	1	1	1	1
1	0	0	1	1	1	0	1
1	0	1	1	1	1	0	1
1	1	0	1	1	1	0	1
1	1	1	1	1	1	1	1

DeMorgan's Theorems

The complement of a product of variables is equal to the sum of the complements of the variables:

$$\underbrace{\overline{AB}}_{\text{complement of a product}} = \underbrace{\overline{A} + \overline{B}}_{\text{sum of the complements}}$$

A	B	$\overline{AB}$	$\overline{A} + \overline{B}$
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0

The complement of a sum of variables is equal to the product of the complements of the variables.

$$\underbrace{\overline{A + B}}_{\text{complement of a sum}} = \underbrace{\overline{A}\,\overline{B}}_{\text{product of the complement}}$$

A	B	$\overline{A + B}$	$\overline{A}\,\overline{B}$
0	0	1	1
0	1	0	0
1	0	0	0
1	1	0	0

Exercise examples

Apply DeMorgan's Theorem to:

1. $\overline{ABC}$
2. $\overline{(A+B+C)D}$
3. $\overline{ABC+DEF}$
4. $\overline{A\overline{B}+\overline{C}D+EF}$
5. $\overline{\overline{A+B\overline{C}}+D(\overline{E+\overline{F}})}$

Summary

- Covered the principles of Boolean algebra
- Familiarised with Boolean terminology
- Learned properties and rules associated with Boolean algebra
- Reviewed additional logic gates (NAND, NOR, XOR, XNOR)